

High end Graphic Cards Test

					Above Rs. 24,500	
Brand	ASUS	ASUS	ASUS	ASUS	ATI	Sapphire
Model	EAH 4870 TOP	EAH 4870	EN GTX 260	EAH 4870 MATRIX	Radeon HD 5870	HD 5870
GPU	Radeon HD 4870	Radeon HD 4870	GeForce GTX 260 216	Radeon HD 4870	Radeon HD 5870	Radeon HD 5870
Price (Rs)	Rs. 22,175	Rs. 22,425	Rs. 22,800	Rs. 24,295	Rs. 24,990	Rs. 26,000
Grand Totals (out of 100)	49.81	46.78	52.33	52.54	71.96	75.61
Features	5.90	5.60	5.95	5.60	3.65	5.70
Performance	43.91	41.18	46.38	46.94	68.31	69.91
Features						
No. of Stream Processors	160 (800 / 5)	160 (800 / 5)	192	160 (800 / 5)	320 (1600 / 5)	320 (1600 / 5)
Core / Shader Clock (MHz)	815 / -	750 / -	576 / 1242	770 / -	850 / -	850 / -
Type of memory / Memory Clock (MHz)	GDDR5 / 3700	GDDR5 / 3600	GDDR3 / 2000	GDDR5 / 3680	GDDR5 / 4800	GDDR5 / 4800
Memory size (MB) / Memory Interface	512 / 256-bit	512 / 256-bit	896 / 448-bit	512 / 256-bit	1024 / 256-bit	1024 / 256-bit
Connects on card	DVI x 2, S-Video x 1	DVI x 2, S-Video x 1	DVI x 2, S-Video x 1	DVI x 2, S-Video x 1	2 x DVI, 1 x HDMI, 1 x Display Port	2 x DVI, 1 x HDMI, 1 x Display Port
In the Box						
Bundled Connects and Cables	1 x 4 pin to 6 pin molex, 1 x DVI-D-Sub converter, Crossfire connect	1 x 4 pin to 6 pin molex, 1 x DVI-D-Sub converter, 1 x S-Video to component converter	Leather mousepad, 1 x 4 pin to 6 pin molex, 1 x DVI-D-Sub converter, 1 x S-Video to component converter	Leather mousepad, 1 x 4 pin to 6 pin molex, 1 x DVI-D-Sub converter, 1 x S-Video to component converter	1 x 4 pin to 6 pin molex, 1 x DVI-D-Sub converter, Crossfire connect	2 x 4 pin to 6 pin molex, 1 x DVI-D-Sub converter
Software and Games	-	-	Alone in the Dark	-	NA	Battlestations Pacific, Dirt 2 voucher
Performance						
Temperature (Idle / load)	51 / 67	49 / 76	44 / 71	50 / 63	48 / 61	48 / 61
3D Mark Vantage (Overall / GPU)	11,277 / 9,780	10,648 / 9,148	10,778 / 8,613	10,637 / 9,353	16,832 / 15,845	16,991 / 15,984
Game Benchmarks						
Crysis Warhead (1680 x 1050, Max / Avg / Min)	26 / 19 / 13	25 / 18 / 13	25 / 19 / 13	24 / 18 / 13	38 / 28 / 21	39 / 29 / 21
Crysis Warhead (1920 x 1200, Max / Avg / Min)	21 / 16 / 11	21 / 15 / 11	20 / 16 / 10	19 / 14 / 9	33 / 24 / 17	34 / 25 / 18
S.T.A.L.K.E.R Clear Sky (Day, 1680 x 1050, Max / Avg / Min)	34.4 / 20 / 10.5	36.8 / 19.1 / 9.7	37.1 / 19.7 / 12.1	40.2 / 21 / 9.2	70.2 / 35.6 / 15.7	71.9 / 36.3 / 16.1
S.T.A.L.K.E.R Clear Sky (Day, 1920 x 1200, Max / Avg / Min)	29.6 / 17.6 / 6.5	28.4 / 16.6 / 6	28.6 / 15.5 / 6.8	32.3 / 17.3 / 6.2	58.6 / 30.2 / 14.2	60.5 / 31.6 / 14.7
Far Cry 2 (1680 x 1050, Max / Avg / Min)	60.12 / 37.03 / 15.62	54.04 / 24.27 / 10.78	83.57 / 47.16 / 34.8	155.3 / 43.63 / 30.43	128.6 / 76.2 / 53.4	144.79 / 76.26 / 55.76
Far Cry 2 (1920 x 1200, Max / Avg / Min)	55.73 / 35.96 / 17.99	52.03 / 29.14 / 10.55	61.36 / 41.97 / 31.14	134.32 / 40.11 / 26.33	138.5 / 66.4 / 46.74	139.04 / 67.08 / 47.22
Unreal Tournament 3 (1680 x 1050 / 1920 x 1200)	151 / 123	143 / 116	133 / 113	147 / 118	194 / 169	195 / 171
Company Of Heroes: Tales Of Valor (1680 x 1050, Max / Avg / Min)	235 / 132.8 / 58.7	219.7 / 124.9 / 57.6	332 / 156.5 / 62.6	253.2 / 130.2 / 53.1	445.3 / 215.3 / 100.2	434.5 / 219 / 101.1
Company Of Heroes: Tales Of Valor (1920 x 1200, Max / Avg / Min)	203.5 / 110 / 47.6	195.2 / 107.2 / 42.8	287 / 129 / 50.4	198.5 / 105.3 / 45.4	478.3 / 241.4 / 110.3	502.4 / 266.3 / 130.7
Visual Quality (Out of 10)						
Crysis Warhead / S.T.A.L.K.E.R Clear Sky	8 / 8	8 / 8	8 / 7.75	8 / 8	9 / 8.5	9 / 8.5



ZOTAC GeForce GTX 285

an incredible 900 MHz showed no signs of overheating – testament to build quality as well as ATI's excellent RV 770 GPU.

Warhead proves a bit too much for these cards. Sure, the maximum fps values are impressive but it isn't representative of gameplay. When one looks at the minimum fps values, we find none of these cards can reach even 20 fps at a modest resolution of 1680 x 1050 pixels

– definitely not for the hardcore gamer who wants a future-proof offering. *Clear Sky* is nearly as bad and at times worse. In fact, owing to the huge levels and viewing distance, the frames dip to unplayable levels when settings are maxed on these cards. *Far Cry 2* is playable, but starts to become a little choppy at a resolution of 1920 x 1200 pixels; however, if one turns off FSAA, things become much better. *Warhead* is one game where FSAA kills nearly all cards. Thankfully, the game looks great

without enabling FSAA and jagged edges aren't as apparent. Older games like *UT 3* and *Tales Of Valor* are no challenge for these cards. A strange trend is noticed when one looks closely at the fps scores. While *Warhead* and *Clear Sky* are equally hard on both ATI and NVIDIA cards, *Far Cry 2*, which is also a new game seems to favour the Radeon HD 4890 over the GTX 260 cards. *UT 3* is also faster on ATI hardware, only *Tales Of Valor* favours NVIDIA.

If you're looking at value for